

GE Drive Systems

INSTRUCTIONS

531X139APMA_G2 APPLICATION CARD

7V11CAC Option Kit

IMPORTANT INFORMATION

CAUTION: To ensure proper operation of the QIII DC Controller, these instructions must be followed for proper installation and set-up of this Option Kit.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

SAFETY PRECAUTIONS

WARNING - SHOCK AND BURN HAZARD:
Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

NOTE: *Read all WARNINGS, CAUTIONS and NOTES in the QIII DC Controller Instruction Book (GEK-45167) prior to installing this Option Kit.*

NOTE: *Read and heed all WARNING, CAUTION, and NOTE labels posted on the equipment prior to installing this Option Kit.*

KIT DESCRIPTION AND CONTENTS

NOTE: *4TB Option Kit (7V11CTB) must be installed prior to installing this kit. When installing Application Card, MODE6 (4TB7) must be jumpred to REF24 (3TB6 or 3TB56).*

DESCRIPTION:

This optional kit provides a combination of analog interfaces between the Control Card and external equipment. These interfaces allow controller to be expanded to handle functions required in various applications. The 531X139APMA_G2 card provides the following optional functions:

- Analog Inputs
- Digital Inputs
- Sixteen Bit Parallel Word Input
- Five Programmable Signal-Level-Detector Relays (SLDs)
- Seven-Bit Digital Fault Status Output
- Three Programmable Status Bit Outputs
- Two Analog Outputs

CONTENTS:

The kit contains the following components (in addition to these instructions):

Component	Part Number	Qty.
Application Card	531X139APMA_G2	1
11PL Conn. Cable (G, C Frames)	36A358208AAG11	1
13PL Conn. Cable (G, C Frames)	36A358208AAG12	1
14PL Conn. Cable (G, C Frames)	36A358208EYG01	1
11PL Conn.Cable (D Frames)	36A358218XAG04	1
13PL Conn. Cable (D Frames)	36A358218XAG06	1
14PL Conn. Cable (D Frames)	36A358208EYG02	1
3PL Conn.Cable (All Frames)	36A358208AAG04	1

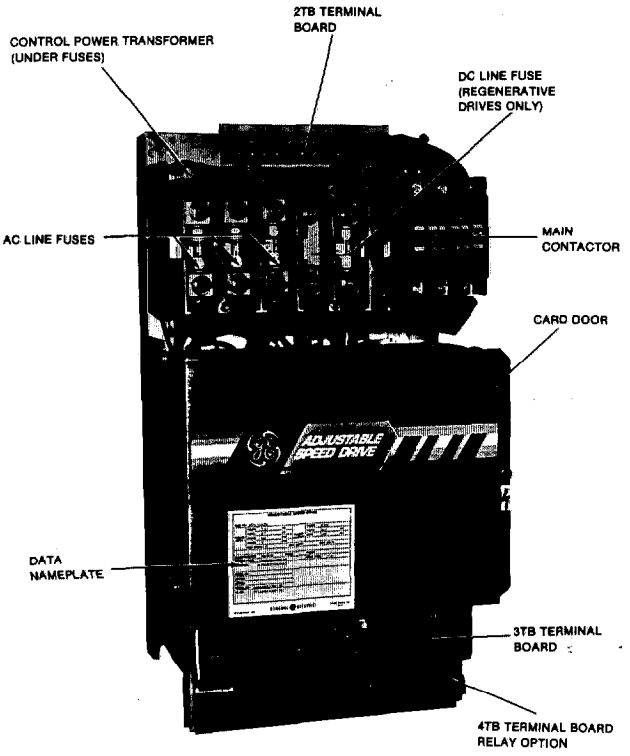


Figure 1. QIII DC CONTROLLER

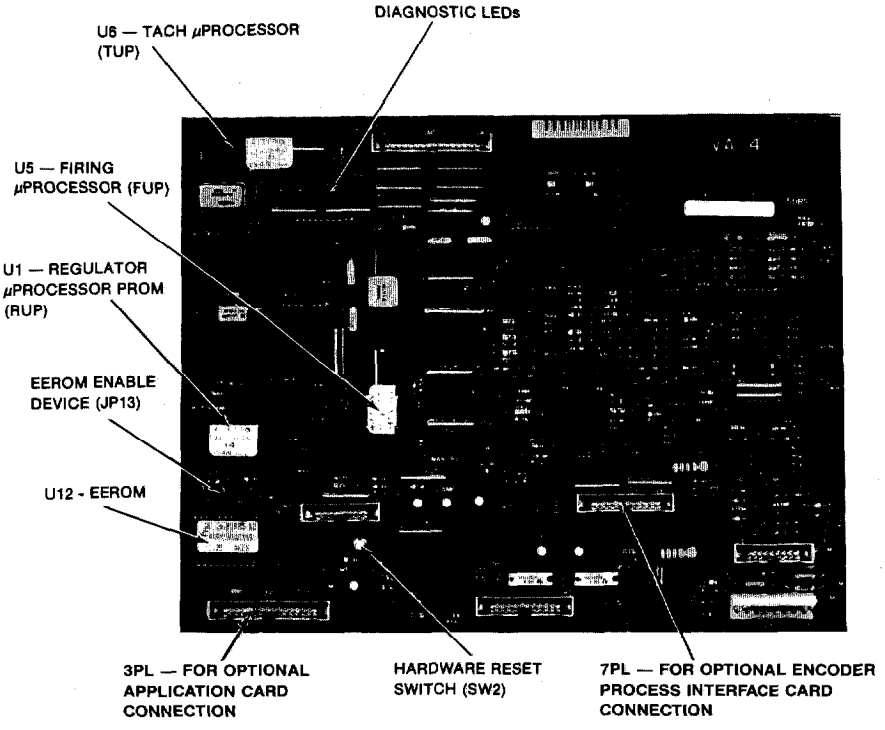


Figure 2. CONTROL CARD

KIT INSTALLATION PROCEDURE

WARNING - SHOCK AND BURN HAZARD:
Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.

1. Verify that power to the controller that the option kit is to be installed is OFF.

2. Loosen thumbscrew (located on right side) and open the controller card door (see Figure 1).

3. Install 3PL ribbon cable on Control card as follows (see Figure 2):
   a. Orient 3PL ribbon cable to its receptacle so that cable extends toward bottom of card rack.
   b. Insert the connector into its receptacle on the Control card and press into fully seated position.

4. Access the card rack mounting location for the Application card as follows (see Figure 3):
   a. Slide each card rack latch bar forward (located on each side of controller).
   b. Lift front card rack upward and allow to tilt forward until it rests in horizontal position.

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

5. Install Application card in card rack as follows (see Figure 4):
   a. Remove card from anti-static bag and orient card with 11PL and 14PL receptacles toward bottom of card rack.
   b. Locate card over the six locating posts on the card rack and press into position until the six card retaining tabs snap into position securing the card.

6. Ensure that 3PL cable (from Control card) is not twisted and insert the connector into its receptacle on the Application card and press into fully seated position (see Figure 4).

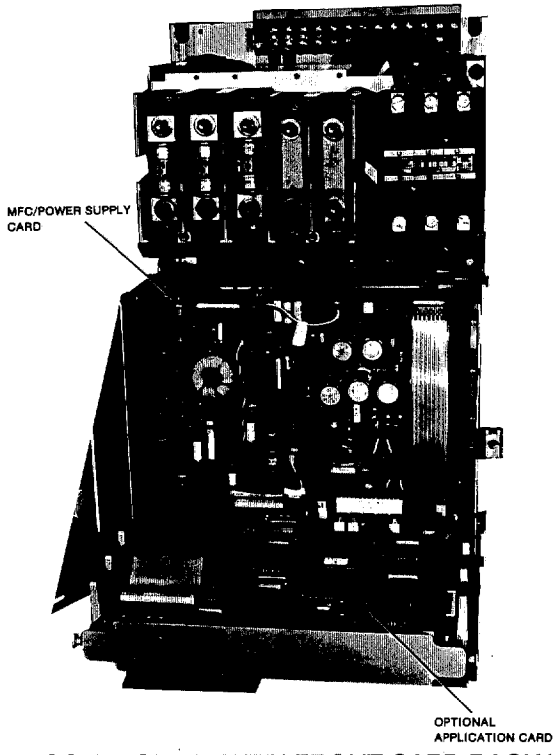


Figure 3. CONTROLLER WITH FRONT CARD RACK LOWERED

7. Install 11PL, 13PL, and 14PL as follows (see Figure 4):
   a. Orient the ribbon cable to its respective receptacle on the Application card with the step on the cable connector towards the retaining clip on the receptacle and press the connector into position.
   b. Repeat above step for the remaining two ribbon cables.
   c. Route all three ribbon cables under bottom of the card rack.

8. Raise the card rack upward to a vertical position and slide rack extensions into slots provided on top of controller until card rack is fully seated in position.

9. Slide the card rack latch bars (located on each side of controller) back.

10. Connect 11PL, 13PL, and 14PL cables to 4TB Terminal Board card as follows (see Figures 5 and 6):

    For "C" and "G" Frame Sizes (see Figure 6):

    a. Verify that cables are not twisted.
    b. Orient the ribbon cable to its respective receptacle on the card with the step on the cable connector towards the retaining clip on the receptacle and press the connector into position.
    c. Repeat above step for the remaining two ribbon cables.

    For "D" Frame Size:

    a. Route all three ribbon cables through opening in bottom of card rack enclosure.
    b. Make 45 degree fold in each ribbon cable and route cables to the right.
    c. Secure the ribbon cables under the three retainers as shown in Figure 5 (release retainer strap, insert the additional ribbon cables, and reattach retainer strap).
    d. Make a mild 45 degree fold in each ribbon cable in order to route cables downward and then another mild 45 degree fold to route cables to the left towards 4TB (see Figure 5).
    e. Secure the ribbon cables under the appropriate retainers as shown in Figure 5 (release retainer strap, insert the additional ribbon cables, and reattach retainer strap).
    f. Make a 45 degree fold in each ribbon cable when it is positioned at its respective receptacle on 4TB (see Figure 6).
    g. Orient each ribbon cable to its respective receptacle on 4TB with the step on the cable connector towards the retaining clip on the receptacle and press each connector into position.

11. Set all jumpers and potentiometers on the newly installed Application card per the Programming Software Package (see Figure 7 for Jumper Location and Table 1 for Jumper Description).

12. Close the controller card rack door and tighten the thumbscrew (located on right side, see Figure 1).

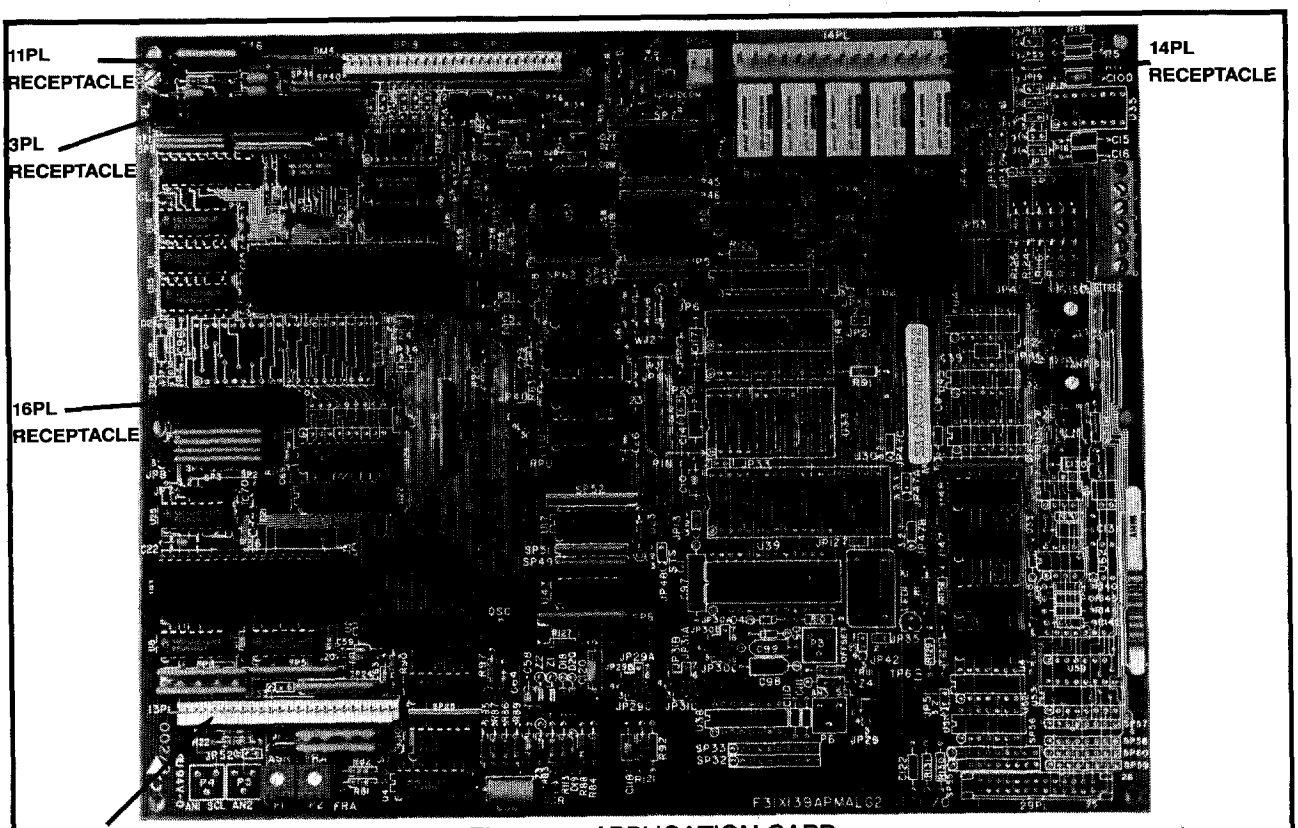


Figure 4. APPLICATION CARD

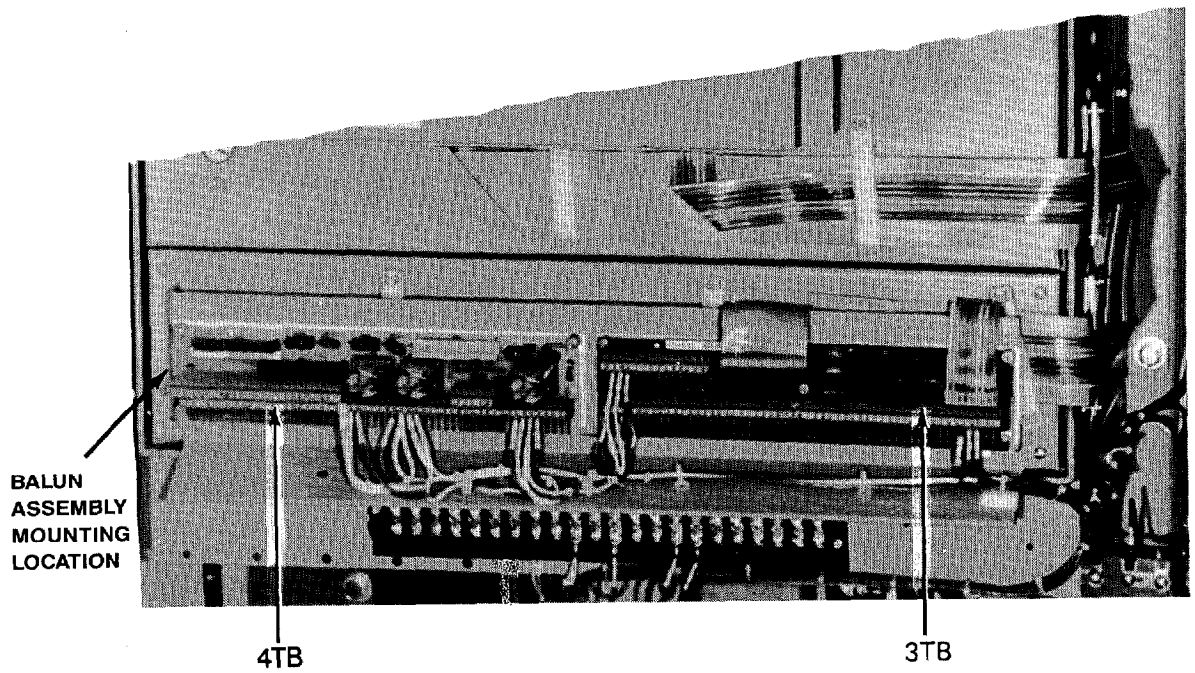


Figure 5. "D" FRAME RIBBON CABLES

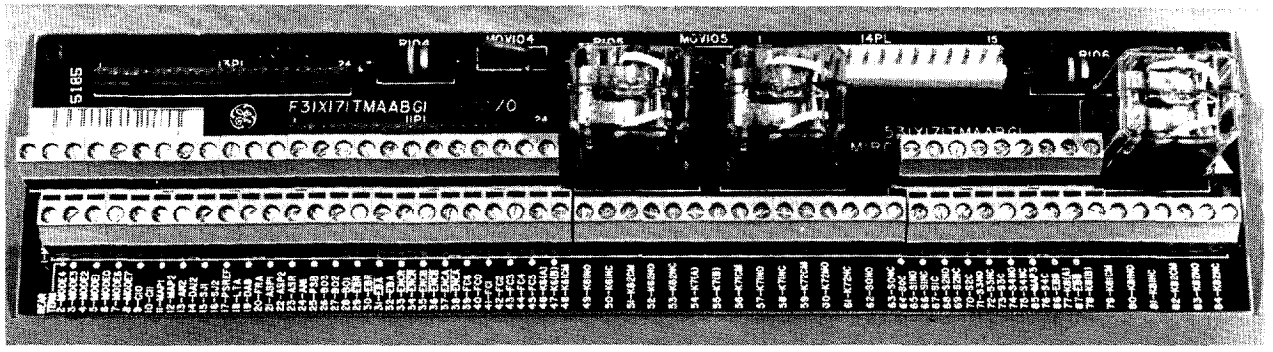


Figure 6. 4TB TERMINAL BOARD (OPTIONAL KIT)

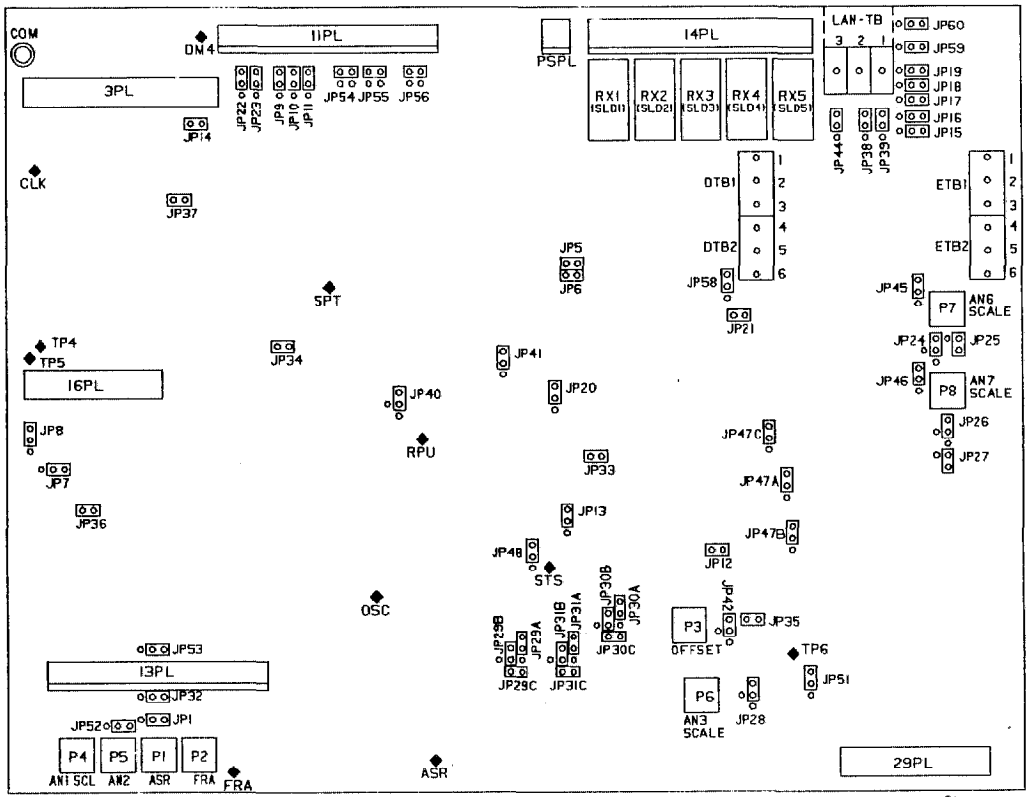


Figure 7. APPLICATION CARD LAYOUT DIAGRAM

TABLE 1: JUMPER DESCRIPTION - 531X139APMA_G2 CARDS

Location	Jumper or Pot	Number	Nomenclature	Description
Application Card	Pot	P1	ASR SCALE	P1 is used in conjunction with JP1 to trim test point ASR to four (4) volts for top speed reference applied to 13PL-23 (4TB23).
		P2	----	Not Used - Set in full CCW position for proper operation.
		P7	----	Not Used - Set in full CCW position for proper operation.
		P8	----	Not Used - Set in full CCW position for proper operation.
	Jumper	JP1		JP1 selects the voltage range for the auto speed reference input (ASR) at 4TB23 (13PL-23). **1-2:** 8 to 15 volts. **2-3:** 5 to 8 volts.
		JP5 JP6		JP5 and JP6 select polarity of the eight logic inputs, MODE 0-MODE 7, at 13PL-1 thru 8 (4TB1 thru 8) and should be set as follows: **1-2 & 3-4:** The control is active low. Inputs are active when pulled below 0.8 volts.
		JP7		JP7 selects whether the sixteen-bit parallel input is latched on the high or low level of the strobe. **1-2:** Data latched on low to high transition of PSB. **2-3:** Data latched on low level of PSB.
		JP8		JP8 selects whether the sixteen-bit parallel input and strobe signals are driven by an active high or low driver. **1-2:** Lines internally pulled up to +5 VDC. Must be driven by active pull-down to common driver. **2-3:** Lines internally pulled down to DCOM. Must be driven by an active pull-up (above 3.5 VDC).
		JP14		JP14 set as follows: **1-2:** Should be in this position for proper operation.
		JP24, JP25 JP26, JP27 JP28		Not Used - set in 1-2 position for proper operation.
		JP29A, B, C JP30A, B, C JP31A, B, C		Not Used - Set as follows for proper operation. **Jumper** — **Position** JP29A, JP30A, JP31A — **6-7:** JP29B, JP30B, JP31B — **1-2:** JP29C, JP30C, JP31C — **3-8:**

Location	Jumper or Pot	Number	Nomenclature	Description
Application Card (continued)	Jumper	JP32		Not Used - set in 1-2 position for proper operation.
		JP36 JP37		JP36 & JP37 set as follows: **1-2:** Should be in this position for proper operation. **Open:** Only for manufacturer's use.
		JP40		Not Used - set in 1-2 position for proper operation.
		JP41		Not Used - set in 1-2 position for proper operation.
		JP45		Not Used - set in 1-2 position for proper operation.
		JP46		Not Used - set in 1-2 position for proper operation.
		JP54		Not Used - set in 1-2 position for proper operation.
		JP55		Not Used - set in 1-2 position for proper operation.
		JP56		Not Used - set in 1-2 position for proper operation.

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